

Repeated Games

Two firms set prices today and meet again tomorrow. A choice today can change how the other firm behaves tomorrow. Fix a normal-form game, called the *stage game*, and play it in periods $t = 0, 1, 2, \dots$. At the end of each period everyone observes the actions played.

In a one-shot game, a strategy tells a player what to do in that game. Here it must also say how to respond to earlier play: for example, cooperate today if everyone has cooperated so far, and defect otherwise.

Definition 1. A *history* h_t records everyone's actions in periods 0 through $t - 1$; h_0 is empty. A *strategy* in the repeated game assigns a stage-game action (or distribution over actions) to every history.

Definition 2. Players have a common *discount factor* $\delta \in (0, 1)$. Player i evaluates a stream of stage payoffs by

$$\sum_{t \geq 0} \delta^t u_i^t,$$

with expectation over any randomization. Stage payoffs are bounded.

A payoff of 1 tomorrow is worth δ today; the same payoff two periods from now is worth δ^2 . Larger δ means that future payoffs matter more. We use unnormalized discounted payoffs: we do not multiply the sum by $1 - \delta$.

What is the value of receiving u every period? Of receiving d now and w every period starting tomorrow?

Subgame Perfection

Nash equilibrium rules out a beneficial unilateral change to a player's whole strategy. But a threat can deter deviations even if the player making it would not want to carry it out. If the threatened history is never reached, changing the action there does not improve that player's payoff from the start of the game.

A threatened punishment must itself be optimal when the time comes to carry it out.

Definition 3. A *subgame* is the repeated game remaining after a history h_t . A strategy profile is a *subgame-perfect equilibrium* (SPE) if it induces a Nash equilibrium after every history, including histories that never occur when everyone follows the profile.

Complete the comparison:

- Nash equilibrium checks deviations starting
- Subgame perfection checks deviations starting

Theorem 1 (One-shot deviation principle). *In this discounted repeated game, a strategy profile is subgame perfect if and only if, after every history, no player can gain by changing their action in one period and following the prescribed strategy thereafter.*

We take the principle as given. “Thereafter” means following the strategy at the history produced by the deviation, including any punishment it triggers. We can therefore compare today’s payoff plus the continuation payoff, instead of checking every possible change to a whole strategy.

Grim Trigger

Payoffs are listed as (row player, column player).

	Column cooperates	Column defects
Row cooperates	(3, 3)	(0, 4)
Row defects	(4, 0)	(1, 1)

Find the stage-game equilibrium and each player's payoff there.

Both players would receive more from (Cooperate, Cooperate). To sustain that outcome, today's action must affect tomorrow's play. Consider the following strategy.

Grim trigger prescribes a target action profile as long as everyone has always played it. After any deviation, it prescribes a fixed Nash equilibrium of the stage game forever.

At any history, compare payoffs from that period onward, treating it as today. Earlier payoffs are fixed, and the common discount factor multiplying all remaining payoffs does not change which continuation is better.

Exercise. For which discount factors does grim trigger sustain (Cooperate, Cooperate) as an SPE? Check both histories with a past deviation and histories without one.

What happens to this incentive comparison as $\delta \rightarrow 1$ and as $\delta \rightarrow 0$?

The argument used the following steps.

Step 1: Find the punishment equilibrium. Check that repeating it makes the punishment credible.

Step 2: Name the target and its per-period payoff. Compute each player's payoff from conforming.

Step 3: Compute the best one-shot deviation. Hold the other players' current actions fixed.

Step 4: Write down $\text{conform} \geq \text{deviate}$. Include the continuation payoff on each side.

Step 5: Solve for δ . Check the condition for every player.

General Threshold (Optional Extension)

Exercise. Now fix a player and let the target payoff be R , the best one-period deviation payoff be D , and the punishment equilibrium payoff be P , where $D > R > P$. Derive the discount-factor threshold.

Holding D and R fixed, what happens when P decreases? Why can't we simply choose any small P we want?

	Column cooperates	Column defects
Row cooperates	(3, 3)	(0, 4)
Row defects	(4, 0)	(1, 1)

Finite Horizons

Suppose instead there are exactly T periods, numbered $0, \dots, T - 1$, and everybody knows this.

Claim 1. If the stage game has a unique Nash equilibrium (including mixed strategies), every SPE of its T -period repetition prescribes that equilibrium after every history.

Why must the same equilibrium be played after every history?

What does the claim imply for ten repetitions of our matrix? Which assumption would fail in a stage game with two Nash equilibria?

Two Periods, Multiple Equilibria

Now consider a different stage game. The row and column players each choose A , B , or C . Payoffs are listed as (row player, column player).

	Column A	Column B	Column C
Row A	(2, 2)	(0, 3)	(0, 0)
Row B	(3, 0)	(1, 1)	(0, 0)
Row C	(0, 0)	(0, 0)	(3, 3)

Find all pure Nash equilibria of the stage game. Is (A, A) a Nash equilibrium?

Exercise. Play this game for exactly two periods, $t = 0, 1$, with payoff $u_i^0 + \delta u_i^1$ and $\delta = 3/4$. Construct an SPE in which both players choose A in period 0. Specify period-1 play after every possible period-0 outcome, and check both players' incentives in both periods.

Which step of the finite-horizon proof no longer works?